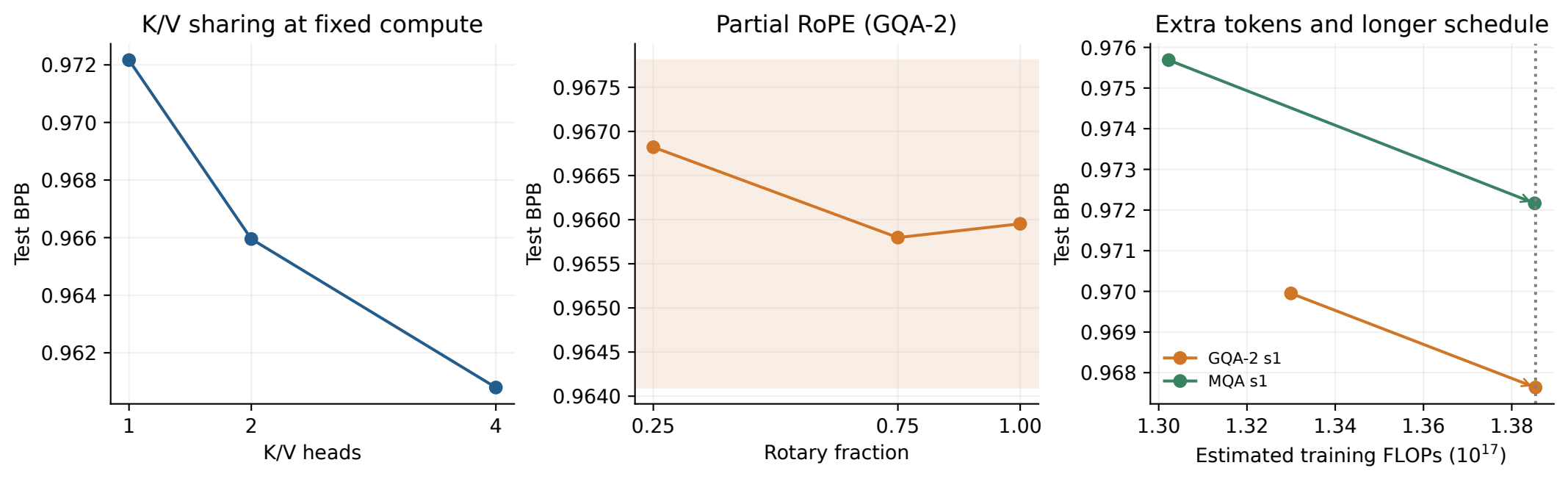
K/V sharing at fixed compute
Test BPB
0.972
0.970
0.968
0.966
0.964
0.962
1
2
4
K/V heads
Partial RoPE (GQA-2)
Test BPB
0.9675
0.9670
0.9665
0.9660
0.9655
0.9650
0.9645
0.9640
0.25
0.75
1.00
Rotary fraction
Extra tokens and longer schedule
Test BPB
0.976
0.975
0.974
0.973
0.972
0.971
0.970
0.969
0.968
GQA-2 s1
MQA s1
1.30
1.32
1.34
1.36
1.38
Estimated training FLOPs (10^{17})